

AI-Powered Resource Allocation & Bench Management

1.0.0 OAS 3.1

/openapi.json

AI-Powered Resource Allocation & Bench Management with RAG

An intelligent system for managing employee resources, project allocations, and bench management using Retrieval-Augmented Generation (RAG) technology.

Key Features:

-  **Resource Management:** Track employees with skills, experience, and availability
-  **Project Management:** Manage projects with requirements and team allocations
-  **Bench Management:** Monitor and manage unallocated resources
-  **AI-Powered Search:** Semantic search using sentence transformers
-  **RAG Queries:** Natural language queries about resources and recommendations
-  **Dashboard:** Analytics on bench utilization and project gaps

Quick Start:

1. Visit `/admin/seed` to populate with sample data
2. Use `/rag/quer` to ask natural language questions
3. Use `/rag/recommend` to get AI-powered resource recommendations
4. Check `/dashboard/stats` for an overview

LLM Configuration:

Set `LLM_PROVIDER` in `.env` to:

- `demo` (default): Returns formatted context without an LLM
- `openai` : Uses OpenAI ChatGPT (requires `OPENAI_API_KEY`)
- `ollama` : Uses local Ollama instance (requires Ollama running)

Resources



GET `/api/v1/resources/` List Resources



POST `/api/v1/resources/` Create Resource



GET `/api/v1/resources/bench` Get Bench Resources



GET `/api/v1/resources/{resource_id}` Get Resource



PUT `/api/v1/resources/{resource_id}` Update Resource



DELETE `/api/v1/resources/{resource_id}` Delete Resource



POST `/api/v1/resources/{resource_id}/allocate` Allocate Resource



POST `/api/v1/resources/{resource_id}/release` Release Resource



Projects



GET `/api/v1/projects/` List Projects



POST `/api/v1/projects/` Create Project



GET `/api/v1/projects/{project_id}` Get Project



PUT `/api/v1/projects/{project_id}` Update Project



DELETE `/api/v1/projects/{project_id}` Delete Project



GET `/api/v1/projects/{project_id}/team` Get Project Team



Allocations



GET /api/v1/allocations/ List Allocations



POST /api/v1/allocations/ Create Allocation



GET /api/v1/allocations/{allocation_id} Get Allocation



DELETE /api/v1/allocations/{allocation_id} Delete Allocation



RAG - AI Queries



POST /api/v1/rag/query Query Resources



POST /api/v1/rag/recommend Recommend Resources



GET /api/v1/rag/search Semantic Search



Dashboard & Analytics



GET /api/v1/dashboard/stats Get Dashboard Stats



GET /api/v1/dashboard/bench-aging Get Bench Aging



GET /api/v1/dashboard/project-gaps Get Project Gaps



GET /api/v1/dashboard/forecast Get Resource Forecast



Admin



POST /api/v1/admin/seed Seed database with sample data



POST /api/v1/admin/reindex Rebuild vector store index



POST	/api/v1/admin/ingest-documents	Upload and index knowledge documents	▼
GET	/api/v1/admin/audit-logs	List audit logs	▼
DELETE	/api/v1/admin/clear	Clear all data	▼
GET	/api/v1/admin/health	Health Check	▼
POST	/api/v1/admin/phase1/prepare-data	Phase 1: collect and store standardized JSON data	▼
POST	/api/v1/admin/phase2/build-rag	Phase 2: build FAISS RAG pipeline	▼
POST	/api/v1/admin/phase3/retrieve	Phase 3: retrieve staffing matches	▼
POST	/api/v1/admin/phase4/recommend	Phase 4: recommendation engine	▼
GET	/api/v1/admin/phase5/dashboard	Phase 5: dashboard metrics	▼

Auth

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POST	/api/v1/auth/login	Login	▼
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Root

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GET	/	Root	▼
GET	/health	Health	▼

Schemas

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AllocationCreate > Expand all object

AllocationInfo > Expand all object

AllocationResponse > Expand all **object**

AllocationSummary > Expand all **object**

Body_ingest_documents_api_v1_admin_ingest_documents_post > Expand all **object**

HTTPValidationError > Expand all **object**

LoginRequest > Expand all **object**

LoginResponse > Expand all **object**

Phase3RetrievalRequest > Expand all **object**

Phase4RecommendationRequest > Expand all **object**

ProjectCreate > Expand all **object**

ProjectListResponse > Expand all **object**

ProjectResponse > Expand all **object**

ProjectUpdate > Expand all **object**

QueryRequest > Expand all **object**

QueryResponse > Expand all **object**

RecommendRequest > Expand all **object**

RecommendResponse > Expand all **object**

ResourceCreate > Expand all **object**

ResourceListResponse > Expand all **object**

ResourceResponse > Expand all **object**

ResourceUpdate > Expand all **object**

ValidationError > Expand all **object**